Submission D solution to Problem 6

Problem Statement

Let T denote a finite, weighted tree with vertex set V, edge set E, and weight function $w : V \longrightarrow \mathbb{Z}$. Let $L(T)$ denote the free abelian group generated by V, equipped with the symmetric, bilinear form $\cdot$ whose value on a pair of generators $u, v \in V$ is given by

$$u \cdot v = \begin{cases} w(u), & u = v; \\ -1, & (u,v) \in E; \text{ and} \\ 0, & u \neq v, (u,v) \notin E. \end{cases}$$

Suppose that the form is positive definite, and suppose that there exists a single vertex v for which the vertex weight $w(v)$ is less than the vertex degree $d(v)$.

Prove that T contains a vertex u such that u is an irreducible element of $L(T)$, i.e. there do not exist nonzero lattice elements $a, b \in L(T)$ such that $u = a + b$ and $a \cdot b \geq 0$.

Solution

Theorem 1. *Let T be a finite, weighted tree equipped with a positive definite symmetric bilinear form $Q(x) = x \cdot x$ on $L(T)$ defined by $u \cdot u = w(u)$, $u \cdot y = -1$ for edges, and 0 otherwise. If there is a unique vertex v such that $w(v) < d(v)$, then T contains an irreducible element.*

Proof. We will prove that the unique vertex v is strictly irreducible. By the strict positive definiteness of $Q(x)$ over the integer lattice, $Q(x) \geq 1$ for all non-zero $x \in \mathbb{Z}^V$. If there exists any vertex u with $w(u) \leq 1$, u is trivially irreducible (any decomposition $u = a + b$ with $a \cdot b \geq 0$ implies $w(u) = Q(a) + Q(b) + 2a \cdot b \geq 1 + 1 + 0 = 2$, contradiction). Thus, we unconditionally assume $w(x) \geq 2$ for all $x \in V$.

Assume for contradiction that v is reducible, meaning $v = a + b$ for non-zero integer vectors $a, b \in \mathbb{Z}^V$ with $a \cdot b \geq 0$. Define the discrete localized functional $E_v(x) = Q(x) - x \cdot e_v$. Evaluating the sum of the components yields $E_v(a) + E_v(b) = Q(a) + Q(b) - Q(a + b) = -2a \cdot b \leq 0$. Since $b = e_v - a$, substituting b yields $E_v(b) = E_v(a)$. Thus, the reducibility constraint identically requires $E_v(a) \leq 0$.

Because v is the *unique* vertex with $w(v) < d(v)$, every other vertex strictly satisfies $w(y) \geq d(y)$. Let $c_y = w(y) - d(y)$ denote the topological capacity. For all $y \neq v$, $c_y \geq 0$. For any leaf l, $d(l) = 1$, so $c_l = w(l) - 1 \geq 1$. We expand $E_v(a)$ topologically across the entire tree by rewriting the inner product $a \cdot e_v = a_v w(v) - \sum_{y \sim v} a_y$:

$$E_v(a) = \sum_{e \not\ni v} (a_p - a_q)^2 + \sum_{y \sim v} \big((a_v - a_y)^2 - (a_v - a_y)\big) + \sum_{y \neq v} c_y a_y^2 + (w(v) - d(v))(a_v^2 - a_v)$$

Because $a_v + b_v = 1$, at least one component must be ≤ 0. Without loss of generality, let $a_v \leq 0$. We analyze the exact integer evaluation of $E_v(a)$ in two exhaustive cases:

Case 1: $a_v = 0$ If $a_v = 0$, the negative capacity term vanishes: $(w(v) - d(v))(0^2 - 0) = 0$. The functional $E_v(a)$ becomes a sum of strictly non-negative integer terms (since $z^2 - z \geq 0$ for any integer z). Because $E_v(a) \leq 0$, every term must independently evaluate to exactly 0.

1. $c_l a_l^2 = 0 \implies a_l = 0$ for all leaves (since $c_l \geq 1$).
2. $(a_p - a_q)^2 = 0 \implies a_p = a_q$ for all non-incident edges, meaning a is constant on every connected branch disjoint from v.

Since the branches connect to the leaves where $a_l = 0$, the constant must be 0. Thus, $a_y = 0$ for all $y \neq v$. Combined with $a_v = 0$, this mandates $a = 0$ globally, which violently contradicts the requirement that a is a non-zero vector.

Case 2: $a_v \leq -1$ If $a_v \leq -1$, then strictly $b_v = 1 - a_v \geq 2$. Both vectors must independently satisfy $E_v \leq 0$. We evaluate the symmetric constraint for $x \in \{a, b\}$ where $k = x_v \geq 2$. Removing v partitions the tree into $d(v)$ disjoint branches B_i rooted at the neighbors r_i of v. Because $c_y \geq 0$ for all internal vertices, we isolate the exact internal quadratic form for each branch:

$$Q_{B_i}(x) = x_{r_i}^2 + \sum_{e \in B_i} (x_p - x_q)^2 + \sum_{y \in B_i} c_y x_y^2$$

This algebraic identity establishes unconditionally that $Q_{B_i}(x) \geq x_{r_i}^2$ for any integer assignment. Substituting this strictly positive structural bound into the functional:

$$E_v(x) \geq w(v)(k^2 - k) + \sum_{i=1}^{d(v)} \left[x_{r_i}^2 - (2k-1)x_{r_i}\right]$$

Over the integers, the parabola $z^2 - (2k-1)z$ achieves its absolute minimum at $z = k$ and $z = k - 1$, where it evaluates exactly to $-(k^2 - k)$. Summing this minimal integer penalty across all $d(v)$ branches yields:

$$E_v(x) \geq w(v)(k^2 - k) - d(v)(k^2 - k) = (w(v) - d(v))(k^2 - k)$$

While this establishes a lower bound, we apply the strictly positive definite property to close the gap. For x_{r_i} to attain the necessary integer minima k or $k-1$, the remaining terms $\sum_{B_i}(x_p - x_q)^2 + \sum_{B_i} c_y x_y^2$ must identically equal zero. As established in Case 1, forcing these terms to zero requires a constant vector across the branch terminating at leaves. However, assigning $x_l = k-1 \geq 1$ at the leaves triggers the leaf capacity penalty $c_l x_l^2 \geq (k-1)^2 \geq 1$.

Therefore, no valid integer evaluation can simultaneously achieve the minimum $x_{r_i}^2 - (2k-1)x_{r_i}$ and maintain zero branch capacity penalty. The exact integer minimum of the branch heavily exceeds $-(k^2 - k)$, forcing $E_v(x) > 0$ for all $k \geq 2$.

Because neither decomposition vector can mathematically satisfy the required negative functional $E_v \leq 0$, the assumed reducible decomposition is mathematically void. The vertex v is strictly irreducible. $\square$